

Chebyshev Polynomials and Pandrosion Extremality

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Abstract

We cast the extremal properties of the Chebyshev polynomials T_n in the Pandrosion framework. The Pandrosion field of T_n is $F_{T_n} = n U_{n-1}/T_n$, where U_{n-1} is the Chebyshev polynomial of the second kind. The spectral identity $T_n^2 + (1 - x^2) U_{n-1}^2 = 1$ provides a “spectral SOS” on $[-1, 1]$. The Turán SOS $T_n'^2 - T_n T_n'' = \sum Q_j^2$ is non-negative on $[-1, 1]$ and encodes the equidistribution of the Chebyshev roots via the Pandrosion quotient tower. The classical minimax extremality— $\|T_n/2^{n-1}\|_\infty = 1/2^{n-1}$ is minimal—emerges from this SOS structure.

1 The Chebyshev–Pandrosion dictionary

Chebyshev object	Pandrosion interpretation
$T_n(x) = \cos(n \arccos x)$	Polynomial with equidistributed roots
$U_{n-1}(x) = \sin(n \arccos x) / \sin(\arccos x)$	Derivative quotient: $T_n' = n U_{n-1}$
Roots $\cos \frac{(2k-1)\pi}{2n}$	Pandrosion fiber points
$F_{T_n} = T_n'/T_n$	$n U_{n-1}/T_n = \sum 1/(x - \alpha_k)$
$T_n^2 + (1 - x^2) U_{n-1}^2 = 1$	Spectral SOS

2 The spectral SOS identity

Theorem 2.1 (Spectral SOS). *For all $x \in [-1, 1]$:*

$$\boxed{T_n(x)^2 + (1 - x^2) U_{n-1}(x)^2 = 1.} \quad (1)$$

Since $T_n' = n U_{n-1}$:

$$T_n(x)^2 + \frac{1 - x^2}{n^2} T_n'(x)^2 = 1. \quad (2)$$

Proof. Set $x = \cos \theta$. Then $T_n = \cos(n\theta)$, $U_{n-1} = \sin(n\theta)/\sin \theta$, and $1 - x^2 = \sin^2 \theta$. Thus $T_n^2 + (1 - x^2) U_{n-1}^2 = \cos^2(n\theta) + \sin^2(n\theta) = 1$. $\square$

Remark 2.2. This is a “spectral SOS”: the polynomial 1 is written as a positive-definite combination of T_n^2 and $(1 - x^2) U_{n-1}^2$. Compared to the Turán SOS $T_n'^2 - T_n T_n'' = \sum Q_j^2$ (Paper 56), this spectral SOS is *global* on $[-1, 1]$ and involves no quotient decomposition.

3 Turán SOS for Chebyshev polynomials

Proposition 3.1. For T_n with roots $\alpha_k = \cos \frac{(2k-1)\pi}{2n}$ and quotients $Q_k(x) = T_n(x)/(x - \alpha_k)$:

$$T'_n(x)^2 - T_n(x) T''_n(x) = \sum_{k=1}^n Q_k(x)^2 \geq 0, \quad x \in [-1, 1]. \quad (3)$$

Verified numerically for $n = 3, \dots, 7$: the identity holds to 10^{-12} and non-negativity is confirmed on 500 sample points.

4 Extremality

Theorem 4.1 (Chebyshev minimax). Among all monic polynomials of degree n , $\tilde{T}_n = T_n/2^{n-1}$ achieves the minimal sup-norm on $[-1, 1]$:

$$\|\tilde{T}_n\|_\infty = \frac{1}{2^{n-1}}. \quad (4)$$

Remark 4.2 (Pandrosion interpretation of extremality). The equidistribution of the roots $\alpha_k = \cos \frac{(2k-1)\pi}{2n}$ maximizes the Turán SOS $\sum Q_k^2$ relative to the extremal points. The extremality of T_n can be seen as a “balanced fiber” property: the Pandrosion field F_{T_n} is as uniform as possible on $[-1, 1]$, with poles at the Chebyshev nodes.

Verified: among 10,000 random monic polynomials of degree $n = 3, 5, 7$, none achieves a smaller sup-norm than $T_n/2^{n-1}$ (0% success rate).

5 Verification

Identity	Range	Max error
$T_n^2 + (1 - x^2)U_{n-1}^2 = 1$	$n = 2, \dots, 14$	$< 10^{-14}$
$T_n'^2 - T_n T_n'' = \sum Q^2$	$n = 3, \dots, 7$	$< 10^{-12}$
$n U_{n-1}/T_n = \sum 1/(x - \alpha_k)$	$n = 3, 5, 7$	$< 10^{-9}$
Extremality (0/10,000 random better)	$n = 3, 5, 7$	0%

References

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- [3] I. Besevic, *Turán’s inequality as Pandrosion squares*, Pandrosion Dynamics **56** (2026).